

## PROBLEMS AND SOLUTIONS.

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[Vol. XIV., July to December, 1870, pp. 17–19.]

**3002.** (Proposed by Matthew Collins, B.A.)—If every two of five circles  $A, B, C, D, E$  touch each other, except  $D$  and  $E$ , prove that the common tangent of  $D$  and  $E$  is just twice as long as it would be if  $D$  and  $E$  touched each other.

*Solution by* PROFESSOR CAYLEY.

Consider the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

foci  $R, S$ ; the coordinates of a point  $U$  on the ellipse may be taken to be  $(a \cos u, b \sin u)$ , and then the distances of this point from the foci will be

$$r = a(1 - e \cos u), \quad s = a(1 + e \cos u).$$

Taking  $k$  arbitrarily, with centre  $R$  describe a circle radius  $a - k$ , with centre  $S$  a circle radius  $a + k$ , and with centre  $U$  a circle radius  $k - ae \cos u$ : say these are the circles  $R, S, U$  respectively; the circle  $U$  will touch each of the circles  $R, S$  (viz. assuming  $ae < k < a$ , so that the foregoing radii are all positive, it will touch the circle  $R$  externally and the circle  $S$  internally).

Considering next a point  $V$ , coordinates  $(a \cos v, b \sin v)$ , and the circle described about this point with the radius  $k - ae \cos v$ , say the circle  $V$ ; this will touch in like manner the circles  $R, S$  respectively. And the circles  $U, V$  may be made to touch each other externally; viz. this will be the case if squared sum of radii = squared distance of centres, or what is the same thing, squared difference of radii + 4 times the product of radii = squared distance of centres; that is,

$$a^2 e^2 (\cos u - \cos v)^2 + 4(k - ae \cos u)(k - ae \cos v) = a^2 (\cos u - \cos v)^2 + b^2 (\sin u - \sin v)^2;$$

or

$$2(k - ae \cos u)(k - ae \cos v) = b^2 \{1 - \cos(u - v)\}.$$

If for a moment we write  $\tan \frac{1}{2}u = x$ ,  $\tan \frac{1}{2}v = y$ , and therefore

$$\cos u = \frac{1 - x^2}{1 + x^2}, \quad \cos v = \frac{1 - y^2}{1 + y^2}, \quad \sin u = \frac{2x}{1 + x^2}, \quad \sin v = \frac{2y}{1 + y^2},$$

$$\cos(u - v) = \frac{(1 - x^2)(1 - y^2) + 4xy}{(1 + x^2)(1 + y^2)}, \quad 1 - \cos(u - v) = \frac{2(x - y)^2}{(1 + x^2)(1 + y^2)},$$

we have

$$\left\{ k - \frac{ae(1 - x^2)}{1 + x^2} \right\} \left\{ k - \frac{ae(1 - y^2)}{1 + y^2} \right\} = \frac{b^2(x - y)^2}{(1 + x^2)(1 + y^2)},$$

or

$$\{k - ae + (k + ae)x^2\} \{k - ae + (k + ae)y^2\} = b^2(x - y)^2,$$

which is readily identified with the circular relation

$$\tan^{-1} y \left( \frac{k + ae}{k - ae} \right)^{\frac{1}{2}} - \tan^{-1} x \left( \frac{k + ae}{k - ae} \right)^{\frac{1}{2}} = \tan^{-1} \left( \frac{k^2 - a^2 e^2}{a^2 - k^2} \right)^{\frac{1}{2}}.$$

Or, what is the same thing, in order that the circles  $U, V$  may touch, the relation between the parameters  $u, v$  must be

$$\tan^{-1} \left\{ \left( \frac{k+ae}{k-ae} \right)^{\frac{1}{2}} \tan \frac{1}{2}v \right\} - \tan^{-1} \left\{ \left( \frac{k+ae}{k-ae} \right)^{\frac{1}{2}} \tan \frac{1}{2}u \right\} = \tan^{-1} \left( \frac{k^2 - a^2e^2}{a^2 - k^2} \right)^{\frac{1}{2}}.$$

Considering in like manner a circle, centre the point  $W$ , coordinates  $(a \cos w, b \sin w)$ , and radius  $k - ae \cos w$ , say the circle  $W$ ; this will, as before, touch the circles  $R, S$ ; and we may make  $W$  touch each of the circles  $U, V$ ; viz. we must have

$$\tan^{-1} \left\{ \left( \frac{k+ae}{k-ae} \right)^{\frac{1}{2}} \tan \frac{1}{2}w \right\} - \tan^{-1} \left\{ \left( \frac{k+ae}{k-ae} \right)^{\frac{1}{2}} \tan \frac{1}{2}v \right\} = \tan^{-1} \left( \frac{k^2 - a^2e^2}{a^2 - k^2} \right)^{\frac{1}{2}},$$

$$\tan^{-1} \left\{ \left( \frac{k+ae}{k-ae} \right)^{\frac{1}{2}} \tan \frac{1}{2}u \right\} - \tan^{-1} \left\{ \left( \frac{k+ae}{k-ae} \right)^{\frac{1}{2}} \tan \frac{1}{2}w \right\} = \tan^{-1} \left( \frac{k^2 - a^2e^2}{a^2 - k^2} \right)^{\frac{1}{2}},$$

where, in the last equation,

$$\tan^{-1} \left\{ \left( \frac{k+ae}{k-ae} \right)^{\frac{1}{2}} \tan \frac{1}{2}u \right\}$$

must be considered as denoting its value in the first equation increased by  $\pi$ . Hence, adding the three equations, we have

$$\pi = 3 \tan^{-1} \left( \frac{k^2 - a^2e^2}{a^2 - k^2} \right)^{\frac{1}{2}},$$

that is,

$$\left( \frac{k^2 - a^2e^2}{a^2 - k^2} \right)^{\frac{1}{2}} = \tan \frac{1}{3}\pi = \sqrt{3};$$

or

$$k^2 - a^2e^2 = 3(a^2 - k^2),$$

that is,

$$3a^2 - 4k^2 + a^2e^2 = 0;$$

viz. this is the condition for the existence of the three circles  $U, V, W$ , each touching the two others, and also the circles  $R, S$ .

The circle  $R$  lies inside the circle  $S$ , and the tangential distance is thus imaginary; but defining it by the equation

$$\text{squared tangential dist.} = \text{squared dist. of centres} - \text{squared sum of radii},$$

the squared tangential distance is

$$= 4a^2e^2 - 4a^2.$$

But if the circles were brought into contact, the distance of the centres would be  $2k$ , and the value of the squared tangential distance  $= 4k^2 - 4a^2$ ; hence, if this be  $=$  one-fourth of the former value, we have

$$4(k^2 - a^2) = a^2e^2 - a^2,$$

that is,

$$3a^2 - 4k^2 + a^2e^2 = 0,$$

the same condition as above. The solution might easily be varied in such wise that the circles  $R, S$  should be external to each other, and therefore the tangential distance real; but the case here considered, where the locus of the centres of the circles  $U, V, W$  is an ellipse, is the more convenient, and may be regarded as the standard case.

**3144.** (Proposed by Professor Cayley.)—If the extremities  $A, A'$  of a given line  $AA'$  describe given lines respectively, show that there is a point rigidly connected with  $AA'$  which describes a circle.

[Vol. XIV., pp. 67, 68.]

**3120.** (Proposed by Professor Cayley.)—To find the equation of the Jacobian of the quadric surfaces through the six points

$$(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1), (1, 1, 1, 1), (\alpha, 0, \gamma, \delta).$$

*Solution by the PROPOSER.*

Writing for shortness

$$a = \beta - \gamma, \quad b = \gamma - \alpha, \quad c = \alpha - \beta, \quad f = \alpha - \delta, \quad g = \beta - \delta, \quad h = \gamma - \delta,$$

(so that  $a + h - g = 0$ , &c.,  $a + b + c = 0$ ,  $af + bg + ch = 0$ ), the six points lie in each of the plane-pairs

$$\begin{aligned} x(hy - gz + aw) &= 0, & y(-hx + fz + bw) &= 0, \\ z(gx - fy + cw) &= 0, & w(-ax - by - cz) &= 0. \end{aligned}$$

We cannot take these as the four quadrics, on account of the identical equation  $0 = 0$ , which is obtained by adding the four equations; but we may take the first three of them for three of the quadrics, and for the fourth quadric the cone, vertex  $(0, 0, 0, 1)$ , which passes through the other five points; viz. this is

$$a\alpha yz + b\beta zx + c\gamma xy = 0.$$

We write therefore

$$\begin{aligned} P &= x(hy - gz + aw), & Q &= y(-hx + fz + bw), \\ R &= z(gx - fy + cw), & S &= a\alpha yz + b\beta zx + c\gamma xy; \end{aligned}$$

and we equate to zero the determinant formed with the derived functions of  $P, Q, R, S$  in regard to the coordinates  $(x, y, z, w)$  respectively. If, for a moment, we write  $A, B, C$  to denote  $bg - ch$ ,  $ch - af$ ,  $af - bg$  respectively, it is easily found that the term containing  $d_x S$  is

$$(b\beta z + c\gamma y)x(-agh, bhf, cfg, abc, -af^2, -gB, hC, aA, b^2g, -c^2h)(x, y, z, w)^2 :$$

the terms containing  $d_y S$  and  $d_z S$  are derived from this by a mere cyclical interchange of the letters  $(x, y, z)$ ,  $(A, B, C)$ ,  $(a, b, c)$ , and  $(f, g, h)$ . Collecting and reducing, it is found that the whole equation divides by  $2abc$ ; and that, omitting this factor, the result is

$$\left. \begin{aligned} &ayz(aw^2 - \delta x^2) + fxw(\beta z^2 - \gamma y^2) \\ &+ bzx(\beta w^2 - \delta y^2) + gyw(\gamma x^2 - \alpha z^2) \\ &+ cxy(\gamma w^2 - \delta z^2) + hzw(\alpha y^2 - \beta x^2) \end{aligned} \right\} = 0,$$

which, substituting for  $a, b, c, f, g, h$  their values, is the required form.

If, in the equation, we write for instance  $x = 0$ , the equation becomes

$$ayzw(hy - gz + aw) = 0;$$

or, the section by the plane is made up of four lines. Calling the given points 1, 2, 3, 4, 5, 6, it thus appears that the surface contains the fifteen lines 12, 13, ..., 56, and also the ten lines 123.456, &c.; in all twenty-five lines. Moreover, since the surface contains the lines 12, 13, 14, 15, 16, it is clear that the point 1 is a node (conical point) on the surface; and the like as to the points 2, 3, 4, 5, 6.

**3249.** (Proposed by Professor Cayley.)—Given on a given conic two quadrangles  $PQRS$  and  $pqrs$ , having the same centres, and such that  $P, p; Q, q; R, r; S, s$  are the corresponding vertices (that is, the four lines  $PQ, RS, pq, rs$  all pass through the same point; and similarly the lines  $PR, QS, pr, qs$ , and the lines  $PS, QR, ps, qr$ ); it is required to show that a conic may be drawn, passing through the points  $p, q, r, s$  and touched at these points by the lines  $pP, qQ, rR, sS$ , respectively.

*Solution by the PROPOSER.*

Taking the centres for the vertices of the fundamental triangle, the equation of the given conic may be taken to be  $x^2 + y^2 + z^2 = 0$ ; and then the coordinates of  $P, Q, R, S$  to be  $(A, B, C), (A, -B, C), (A, B, -C), (A, -B, -C)$  respectively, where  $A^2 + B^2 + C^2 = 0$ ; and those of  $p, q, r, s$  to be  $(\alpha, \beta, \gamma), (\alpha, -\beta, \gamma), (\alpha, \beta, -\gamma), (\alpha, -\beta, -\gamma)$  respectively, where  $\alpha^2 + \beta^2 + \gamma^2 = 0$ . The required conic, assuming it to exist, will be given by an equation of the form  $lx^2 + my^2 + nz^2 = 0$ . This must pass through the point  $(\alpha, \beta, \gamma)$ , and the tangent at this point must be

$$x(B\gamma - C\beta) + y(C\alpha - A\gamma) + z(A\beta - B\alpha) = 0;$$

that is, we must have  $l\alpha^2 + m\beta^2 + n\gamma^2 = 0$ , and

$$l\alpha : m\beta : n\gamma = B\gamma - C\beta : C\alpha - A\gamma : A\beta - B\alpha.$$

The first condition is obviously included in the second; and the second condition remains unaltered if we reverse the signs of  $B, \beta$ , or of  $C, \gamma$ , or of  $B, \beta$  and  $C, \gamma$ . Hence the conic passing through  $p$ , and touched at this point by  $pP$ , will also pass through the points  $q, r, s$ , and be touched at these points by the lines  $qQ, rR, sS$ , respectively; that is, the equation of the required conic is

$$\frac{B\gamma - C\beta}{\alpha}x^2 + \frac{C\alpha - A\gamma}{\beta}y^2 + \frac{A\beta - B\alpha}{\gamma}z^2 = 0;$$

or, what is the same thing,

$$\begin{vmatrix} B\gamma x^2 & \gamma\alpha y^2 & \alpha\beta z^2 \\ A & B & C \\ \alpha & \beta & \gamma \end{vmatrix} = 0.$$

[Vol. XV., January to June, 1871, pp. 17–20.]

**3206.** (Proposed by Professor Cayley.)—In how many geometrically distinct ways can nine points lie in nine lines, each through three points?

**3278.** (Proposed by Professor Cayley.)—It is required, with nine numbers each taken three times, to form nine triads containing twenty-seven distinct duads (or, what is the same thing, no duad twice), and to find in how many essentially distinct ways this can be done.

*Solution by the PROPOSER.*

Let the numbers be 1, 2, 3, 4, 5, 6, 7, 8, 9. Any number, say 1, enters into three triads, no two of which have any number in common. We may take these triads to be 123, 145, 167. There remain the two numbers 8, 9; and these are, or are not, a duad of the system.

*First Case.*—8 and 9 a duad. In the triad which contains 89, the remaining number cannot be 1; it must therefore be one of the numbers 2, 3; 4, 5; 6, 7; and it is quite immaterial which; the triad may therefore be taken to be 289. There is one other triad containing 2, the remaining two numbers thereof being taken from the numbers 4, 5; 6, 7. They cannot be 4, 5

or 6, 7; and it is indifferent whether they are taken to be 4, 6; 4, 7; 5, 6, or 5, 7: the triad is taken to be 247. We have thus the triads

$$123, 145, 167, 289, 247;$$

and we require two triads containing 8 and two triads containing 9. These must be made up with the numbers 3, 4, 5, 6, 7: but as no one of them can contain 47, it follows that, of the two pairs which contain 8 and 9 respectively, one pair must be made up with 3, 5, 6, 7, and the other pair with 3, 5, 6, 4; say, the pairs which contain 8 are made up with 3, 5, 6, 7, and those which contain 9 are made up with 3, 5, 6, 4 (since obviously no distinct case would arise by the interchange of the numbers 8, 9). The triads which contain 8 must contain each of the numbers 3, 5, 6, 7, and they cannot be 835, 867, since we have 67 in the triad 167; similarly the triads which contain 9 must contain each of the numbers 3, 5, 6, 4, and they cannot be 945, 936, since we have 45 in 145. Hence the triads can only be

$$\begin{array}{cc|cc} 836, & 857 & 934, & 956, \\ 837, & 856 & 935, & 946; \end{array}$$

and clearly the top row of 8 must combine with the top row of 9, and the bottom row of 8 with the bottom row of 9; that is, the system of the nine triads is

$$123, 145, 167, 289, 247,$$

in combination with

$$836, 857, 934, 956,$$

or else in combination with

$$837, 856, 935, 946.$$

These are really systems of the same form, that is, each of them is of the form

$$\begin{array}{c|c|c} BC\alpha & \beta\gamma a & bcC \\ CA\beta & \gamma\alpha b & caA \\ AB\gamma & \alpha\beta c & abB; \end{array}$$

viz. in the first and second systems respectively we have

| <i>A</i> | <i>B</i> | <i>C</i> | $\alpha$ | $\beta$ | $\gamma$ | <i>a</i> | <i>b</i> | <i>c</i>           |
|----------|----------|----------|----------|---------|----------|----------|----------|--------------------|
| 6        | 1        | 3        | 2        | 8       | 7        | 5        | 4        | 9 (First system),  |
| 5        | 1        | 3        | 2        | 9       | 4        | 6        | 7        | 8 (Second system). |

as one out of many ways of effecting the identification. Observe that there is not in the system any triad of triads containing all the numbers. It thus appears that 8, 9, a duad, gives only a single form of the system.

*Cor.*—It is possible to find in a plane nine points such that the points belonging to the same triad lie in lines. The nine points are, in fact, on a cubic curve; and the figure is that belonging to a theorem of Prof. Sylvester's, according to which it is possible to find on a cubic curve a system of points 1, 2, 4, 5, 7, 8, &c., (a series of numbers not divisible by 3), such that for any triad (such as 145) where the sum of the numbers, one taken negatively, = 0, the three points are in lines; and so also that, if two of the points become identical, in the figure 13 = 14, then there is not any new point, but the preceding points are indefinitely repeated; thus, 2, 14, 16 being in lines, and 14 being = 13, 16 must be = 11, and so on.

*Second and Third Cases.*—8 and 9 do not form a duad. There are thus three triads composed of 8 with (2, 3; 4, 5; 6, 7), and three triads composed of 9 with (2, 3; 4, 5; 6, 7). If with these numbers (2, 3; 4, 5; 6, 7) we form all the arrangements of three duads other than those which contain all or any of the duads 23, 45, 67, there are the eight arrangements

$$\begin{array}{ll} A = 24, 37, 56, & E = 26, 35, 47, \\ B = 24, 36, 57, & F = 26, 34, 57, \\ C = 25, 36, 47, & G = 27, 34, 56, \\ D = 25, 37, 46, & H = 27, 35, 46, \end{array}$$

where  $A$  has a duad in common with  $B$ , with  $D$ , and with  $G$ ; but it has no duad in common with  $C, E, F$ , or  $H$ . We have thus the sixteen pairs

$$\begin{array}{cccc} AC, & AE, & AF, & AH, \\ BD, & BE, & BG, & BH, \\ CF, & CG, & GH, & \\ DE, & DF, & DG, & \\ EG, & FH, & & \end{array}$$

where each pair contains six different duads.

Combining  $AC$  with 8, 9, we have the triads 8(24, 37, 56) and 9(24, 36, 57), that is, the triads

$$824, 837, 856 : \quad 924, 936, 957,$$

which, with the original three triads 123, 145, 167, form a system of nine triads; 8 and 9 might, of course, be interchanged, but no essentially distinct system would arise thereby. Hence we have a system of nine triads by combining the original three triads 123, 145, 167, with any one of the sixteen pairs  $AC, AE$ , &c. But it is sufficient to consider the combinations of the three triads with each of the pairs  $AC, AE, AF, AH$ ; in fact, these are the only systems which contain the triad 824; and since there is no distinction between the two pairs 4, 5 and 6, 7, or between the two numbers of the same pair, it is allowable to take 824 as a triad of the system. Hence

*Second Case.*—The system consists of the three triads combined with  $AE$ ; viz. it is

$$123, 145, 167 : \quad 824, 837, 856 : \quad 926, 935, 947 :$$

which, it is to be observed, consists of three triads of triads, each triad of triads containing all the nine numbers; viz. the system is

$$123, 479, 568 : \quad 145, 269, 378 : \quad 167, 248, 359.$$

*Cor.*—We may have nine points such that the points belonging to the same triad lie in lines, viz. the figure is that of Pascal's hexagon when the conic is a line-pair.

*Third Case.*—Combining the three triads with  $AC, AF$ , or  $AH$ , it is readily seen that we obtain in each case a system of the form

$$\begin{array}{lll} A\alpha\alpha', & A\beta\gamma, & A\beta'\gamma', \\ B\beta\beta', & B\gamma\alpha, & B\gamma'\alpha', \\ C\gamma\gamma', & C\alpha\beta, & C\alpha'\beta', \end{array}$$

viz. in the case where the pair is  $AC$ ; that is, the system is

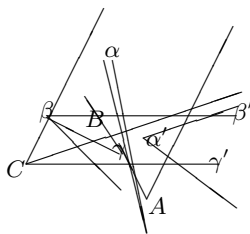
$$123, 145, 167 : \quad 824, 837, 856 : \quad 925, 936, 947$$

and in the cases where the pair is  $AF$  or  $AH$ , the identifications may be taken to be

$$\begin{array}{cccccccccc} A & B & C & \alpha & \beta & \gamma & \alpha' & \beta' & \gamma' & \\ 9 & 8 & 1 & 4 & 5 & 2 & 7 & 6 & 3 & (AC), \\ 9 & 8 & 1 & 2 & 3 & 4 & 6 & 7 & 5 & (AF), \\ 9 & 8 & 1 & 5 & 4 & 6 & 3 & 2 & 7 & (AH). \end{array}$$

Observe that there is in the system a single triad of triads  $A\alpha\alpha', B\beta\beta', C\gamma\gamma'$ , containing all the numbers; viz. for the system with  $AC$ , this is 123, 856, 947; for the system with  $AF$ , it is 145, 837, 926; and for the system with  $AH$ , it is 167, 824, 935.

*Cor.*—It is possible to find a system of nine points such that the points belonging to the same triad lie in lines. Such a figure is this:—



The solution shows that these are the only systems of nine points satisfying the prescribed conditions.

[Vol. XV., pp. 66, 67.]

**3329.** (Proposed by Professor Cayley.)—It is required to show that every permutation of 12345 can be produced by means of the cyclical substitution (12345), and the interchange (12).

*Solution by the PROPOSER.*

It is sufficient to show that the interchanges (13), (14), (15) can be so produced; for then, with the interchanges (12), (13), (14), (15), we can, by at most two such interchanges, bring any number into any place.

Writing  $P = (12345)$ ,  $\alpha = (12)$ , we have

$$(12) = \alpha,$$

$$(13) = \alpha P \alpha P^4 \alpha,$$

$$(14) = \alpha P \alpha P^4 \alpha P^2 \alpha P^3 \alpha P \alpha P^4 \alpha,$$

$$(15) = P^3 \alpha P,$$

as can be at once verified; and the theorem is thus proved.

I remark that, starting with any two or more substitutions, and combining them in every possible manner (each of them being repeatable an indefinite number of times), we obtain a “group”; viz. this is either (as in the problem proposed) the system of all the substitutions (or say the entire group), or else it is a system the number of whose terms is a submultiple of the whole number of substitutions. The interesting question is, to determine those two or more substitutions, which, by their combination as above, do *not* give the entire group; for in this way we should arrive at all the forms of a submultiple group.

[Vol. XV., p. 80.]

**3356.** (Proposed by Professor Cayley.)—If the roots  $(\alpha, \beta, \gamma, \delta)$  of the equation

$$(a, b, c, d, e)(u, 1)^4 = 0$$

are no two of them equal; and if there exist unequal magnitudes  $\theta$  and  $\phi$ , such that

$$(\theta + \alpha)^4 : (\theta + \beta)^4 : (\theta + \gamma)^4 : (\theta + \delta)^4 = (\phi + \alpha)^4 : (\phi + \beta)^4 : (\phi + \gamma)^4 : (\phi + \delta)^4;$$

show that the cubinvariant

$$ace - ad^2 - b^2e - c^3 + 2bcd = 0;$$

and find the values of  $\theta, \phi$ .

[Vol. XVI., June to December, 1871, p. 65.]

**3507.** (Proposed by Professor Cayley.)—Show that, for the quadric cones which pass through six given points, the locus of the vertices is a quartic surface having upon it twenty-five right lines; and, thence or otherwise, that for the quadric cones passing through seven given points the locus of the vertices is a sextic curve.

[Vol. XVI., p. 90.]

**3536.** (Proposed by Professor Cayley.)—A particle describes an ellipse under the simultaneous action of given central forces, each varying as (distance)<sup>-2</sup>, at the two foci respectively: find the differential relation between the time and the excentric anomaly.

[Vol. XVII., January to June, 1872, p. 35.]

**3591.** (Proposed by Professor Cayley.)—If in a plane  $A, B, C, D$  are fixed points and  $P$  a variable point, find the linear relation

$$\alpha.PAB + \beta.PBC + \gamma.PCD + \delta.PDA = 0,$$

which connects the areas of the triangles  $PAB$ , &c.

[Vol. XVII., p. 49.]

**2652.** (Proposed by Professor Cayley.)—Find the differential equation of the parallel surfaces of an ellipsoid.